COURSE

AQA A-level Business (7138)

Operations management: inventory

3.2 Unit: Managing people and operations · 3.2.2 Operations management: inventory

Revision for 3.2 Unit: Managing people and operations: each firm works it out on its own deals.

Class	Topic
AQA A-level Business, new specification (7138)	**Operations management: inventory**

THE LESSON

Setup					3 min
Round 1					Decisions 3 min · Deals 5 min · Topic tasks 4 min
Firm decisions	**Deals**	TASK 1 **Stock holding cost**	TASK 2 **Risks of JIT**		
Round 2					Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min
Firm decisions	**Deals**	TASK 3 **Rank stock turnover**	TASK 4 **Stock turnover**	**B2B deal**	
Round 3					Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min
Firm decisions	**Deals**	TASK 5 **Cut or raise buffer**	TASK 6 **Weeks until reorder**	**B2B deal**	
Final questions					3 min

Fits a 30-minute lesson setting; longer settings add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments FROM ROUND 2	Buy property or funds. The value can rise or fall.
Operations	**Seasonal stock**	Order the season's cars in advance.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

Cars wait in your yard.

Cost of holding the cars?

WORK OUT THE NUMBER · EXAMPLE FIGURES

You hold for them	**10 cars**
Holding cost per car a week	**£90**
Weeks you hold them	**6 weeks**

METHOD

Holding cost = cars × cost per car a week × weeks

10 × £90 × 6 = £5,400

Why: Stock that waits costs £900 every week.

Task 2

ROUND 1

You move to just-in-time stock.

Which two risks rise most?

PICK TWO

- ✔ **A late delivery stops sales**
- ✔ **Fewer bulk-buying discounts**
- ○ Higher storage costs
- ○ More cars go out of date
- ○ More cash tied up in stock

Why: JIT frees cash and space, but depends on every delivery and buys in small lots.

Task 3

ROUND 2

Three dealers: cost of cars sold in a year, and average stock.

Order by stock turnover, fastest first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **£450,000 sold, £50,000 stock**
2. **£600,000 sold, £100,000 stock**
3. **£800,000 sold, £200,000 stock**

Why: Turnover: 9, 6 and 4 times a year. The biggest dealer is the slowest.

Task 4

ROUND 2

One model, one year.

Stock turnover, times a year? Use average stock.

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Cost of cars sold in a year	**£167,400**
Stock at start of year	**£15,420**
Stock at end of year	**£21,780**

METHOD

Turnover = cost of cars sold ÷ average stock

(£15,420 + £21,780) ÷ 2 = £18,600; £167,400 ÷ £18,600 = 9

Why: The stock is sold and replaced 9 times a year.

Task 5

ROUND 3

Deliveries now come a week late. A lost sale costs £1,500; one more car in stock costs £80 a week.

What should you do with buffer stock, and why?

PICK A OR B AND GIVE A REASON

- ○ Cut it to save cash
- ✔ **Raise it**

Why: A lost sale costs far more than a week's holding cost, so a bigger buffer pays.

Task 6

ROUND 3

When do you order next?

Weeks until you reorder?

WORK OUT THE NUMBER · EXAMPLE FIGURES

They buy each week	**4 cars**
Supplier delivers in	**2 weeks**
Buffer, of lead-time sales	**50%**
In stock	**28 cars**

METHOD

Weeks = (stock – reorder level) ÷ weekly sales; buffer = half of lead-time sales

Lead-time sales 8, buffer 4, level 12; (28 − 12) ÷ 4 = 4

Why: A buffer tied to lead time grows when deliveries are slow.

Final questions

6
questions for each pupil

109
Level 3 Business questions

3
of them on operations management: inventory

The game draws 6 of the 109 Level 3 questions; these 3 are about operations management: inventory.

1 What is the central trade-off when a dealership adopts just-in-time delivery for common parts?
Lower holding cost, more supply risk.

2 A workshop uses just-in-time parts supply. Its main supplier shuts for two weeks after a fire. What is the most likely effect?
No buffer, so jobs stop at once.

3 Your only parts supplier's lead time rises from two days to nine. Apart from ordering earlier, what else must change?
More stock must be held, tying up cash and space.